

# SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

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ABSTRACT. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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## 1. AUXILIARY LEMMAS

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful

**Definition 1.1** (Multifold sums of powers). *For non-negative integers  $r, n, m$*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for falling factorials

**Definition 1.2** (Falling factorials). *For integers  $x, n$*

$$(n)_k = \begin{cases} 1, & \text{if } k = 0, \\ n(n-1)(n-2) \cdots (n-k+1) = \prod_{j=0}^{k-1} (n-j), & \text{if } k > 0, \\ 0, & \text{else.} \end{cases}$$

We utilize the following notation for central factorials

**Definition 1.3** (Central factorials). *For integers  $n, k$*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where  $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=0}^{k-2} \left(n + \frac{k}{2} - 1 - j\right)$  are falling factorials.

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [?].

**Lemma 1.4** (Central Newton's formula). *For a function  $f$  defined on integers and arbitrary integers  $x, t$*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where  $\delta^k f(t)$  denotes the  $k$ -th central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right),$$

and  $(x-t)^{[k]}$  are central factorials defined by (1.3).

*Proof.* See [?, p. 462], and [?, section 2, eq. (19)], and [?, section 6, eq. (21)]. □

We have

**Lemma 1.5** (Newton's formula for powers). *For non-negative integers  $n, m$ , and an arbitrary integer  $t$ ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

*Proof.* Special case of Lemma (1.4) for  $f(n) = n^m$ . □

We have

**Lemma 1.6** (Central factorials binomial form). *For integers  $n$  and  $k \neq 0$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

*Proof.* We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k(k-1)!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials  $\frac{(x)_n}{n!} = \binom{x}{n}$ . □

We have

**Lemma 1.7** (Generalized hockey-stick identity). *For arbitrary integers  $a, b$  and  $j$ ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

*Proof.* We have,  $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right)$ . By hockey-stick identity  $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$  yields,

$$\sum_{k=a}^b \binom{k}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof.  $\square$

We have

**Lemma 1.8** (Centered hockey-stick identity). *For integer  $n \geq 1$ , and arbitrary integers  $t, k, r$*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

*Proof.* By setting  $a = 1 - t + \frac{k}{2} + r$ , and  $b = n - t + \frac{k}{2} + r$  into Lemma (1.7) yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof.  $\square$

We have

**Lemma 1.9** (Multifold sums of zero powers). *For non-negative integers  $r, n$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

*Proof.*

- (1) Let  $r = 0$ , then we have  $\Sigma^0 n^0 = 1$ , by definition (1.1).
- (2) Let  $r = 1$ , then we have  $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$ .
- (3) Let  $r = 2$ , then we have  $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$ .
- (4) Let  $r = 3$ , then we have  $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$ .

(5) Similarly, for arbitrary integer  $r$ , by hockey-stick identity  $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$ , yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

This completes the proof.  $\square$

We follow the convention

**Convention 1.10.** *For all  $x$*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention  $x^0 = 1$  for all  $x$ , including zero, in Concrete Mathematics [?, p. 162], and in [?].

## 2. INTRODUCTION

In this manuscript, we derive formulas for sums of powers by combining central Newton's interpolation formula for powers (1.5) with hockey-stick identity (1.8). We implement the algorithm for finding closed forms of sums of powers of integers, proposed in [?, Alg. (11.2)]. More precisely, the main idea is to determine binomial basis of Newton's formula, in terms of variants of difference operators, for example

$$\left\{ \begin{array}{l} f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{d^k f(a)}{dx^k}. \end{array} \right.$$

We may observe that each variation of Newton's formula above has its binomial basis

$$\left\{ \begin{array}{l} \frac{(x)_n}{n!} = \frac{1}{n!} x(x-1)(x-2) \cdots (x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} = \frac{1}{n!} x(x+1)(x+2) \cdots (x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} = \frac{1}{n!} \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = \frac{x}{n} \binom{x+\frac{n}{2}-1}{n-1}. \end{array} \right.$$

This allows us to successfully combine Newton's formula with hockey-stick type identity, to find closed forms for sums of powers of integers. Thus, we focus on closed forms of powers sums, by utilizing central Newton's formula, and hockey-stick identity (1.8).

### 3. MAIN RESULTS

### 4. CONCLUSIONS

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

## MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

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| Mathematica Function                                     | Validates / Prints  |
|----------------------------------------------------------|---------------------|
| <code>ValidateBackwardNewtonsFormulaForPowers.txt</code> | Validates Lem. (??) |

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